# Agenda

01 **Framework**
Inputs, outputs, and the process for managing sustainable advantage

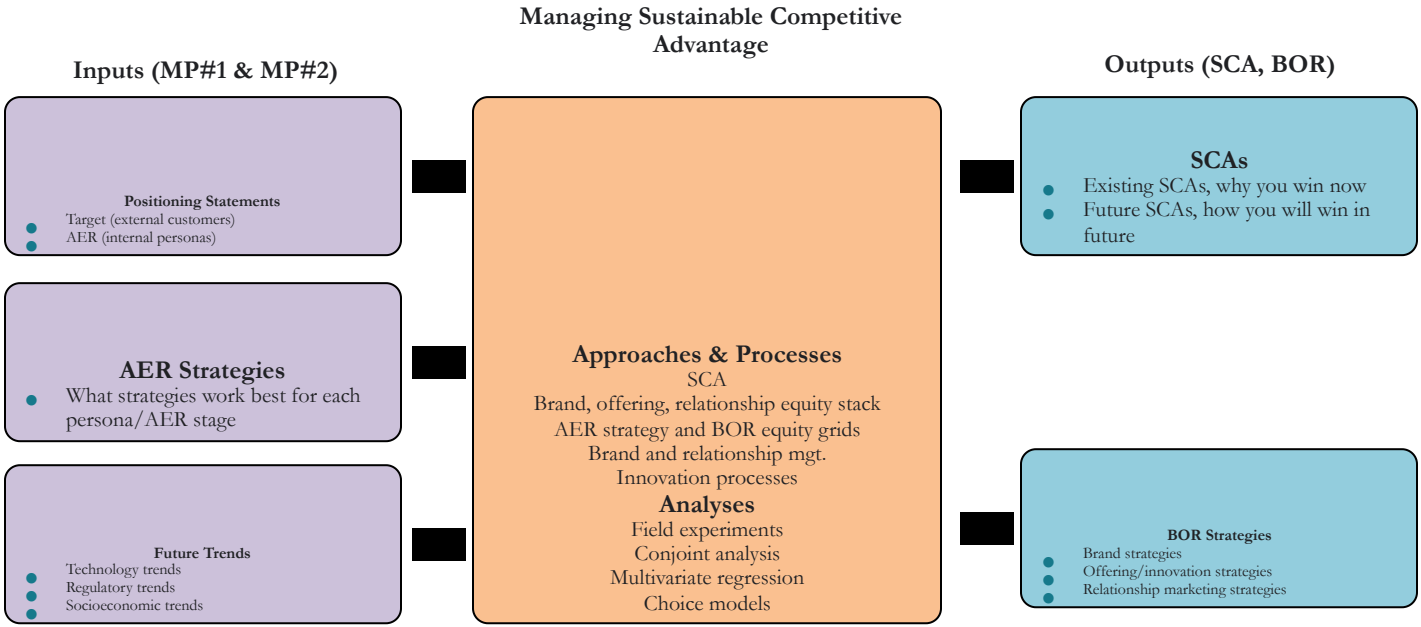
02 **AI Frontier**
Inferring dynamic market structure from social-engagement embeddings

03 **Takeaways**
Connecting customer value, competitive response, and defensibility

# Framework for Managing Sustainable Competitive Advantage

- The organizing framework for managing sustainable competitive advantages (SCA) integrates the preceding approaches and analyses
- Three key inputs:
  - → Positioning statements
  - → AER strategies
  - → Future trends
- Two outputs:
  - → SCAs
  - → BOR strategies

# Marketing Principle #3: All Competitors React → Managing Sustainable Competitive Advantage



# Inputs to the Sustainable Competitive Advantage Framework

- Positioning statements – answers three key questions:
  - → **Who** customers are
  - → **What** set of needs the product or service fulfills
  - → **Why** this product/service is the best option to satisfy customer needs (relative to competition)
- AER strategies – provide key guidance into how a firm should invest to acquire and keep customers
- Future trends – long-term technology, regulatory, and socioeconomic trends, which clearly can disrupt any organization's SCAs

# Example: Brussels Airlines (Belgium)

- Brussels Airlines, Belgium's national carrier, emphasizes the importance of adapting its marketing model with the changing technological trends
- "We have all got to digitally transform to provide the tools to our guests that can help them get through the whole air travel experience. We need to allow them to book online and do everything they need to do from their mobile devices." -Simon Lamkin, CIO

# Outputs of the Sustainable Competitive Advantage Framework

- Descriptions of the firm's SCA now and in the future – needs to clearly address the three conditions for SCA:
  - → Customers care about it
  - → The firm does it better than competitors
  - → It is hard to duplicate or substitute

- Detailed BOR strategies – aggregate and reorganize each targeted customer segment and persona according to its needs; the most effective strategies across time (accounting for customer dynamics) are in the brand, offering, and relationship categories

- Both these outputs (SCA and BOR strategies) represent aggregations of insights gained from more fine-grained analyses, combined and reorganized to support effective macro-level decision making

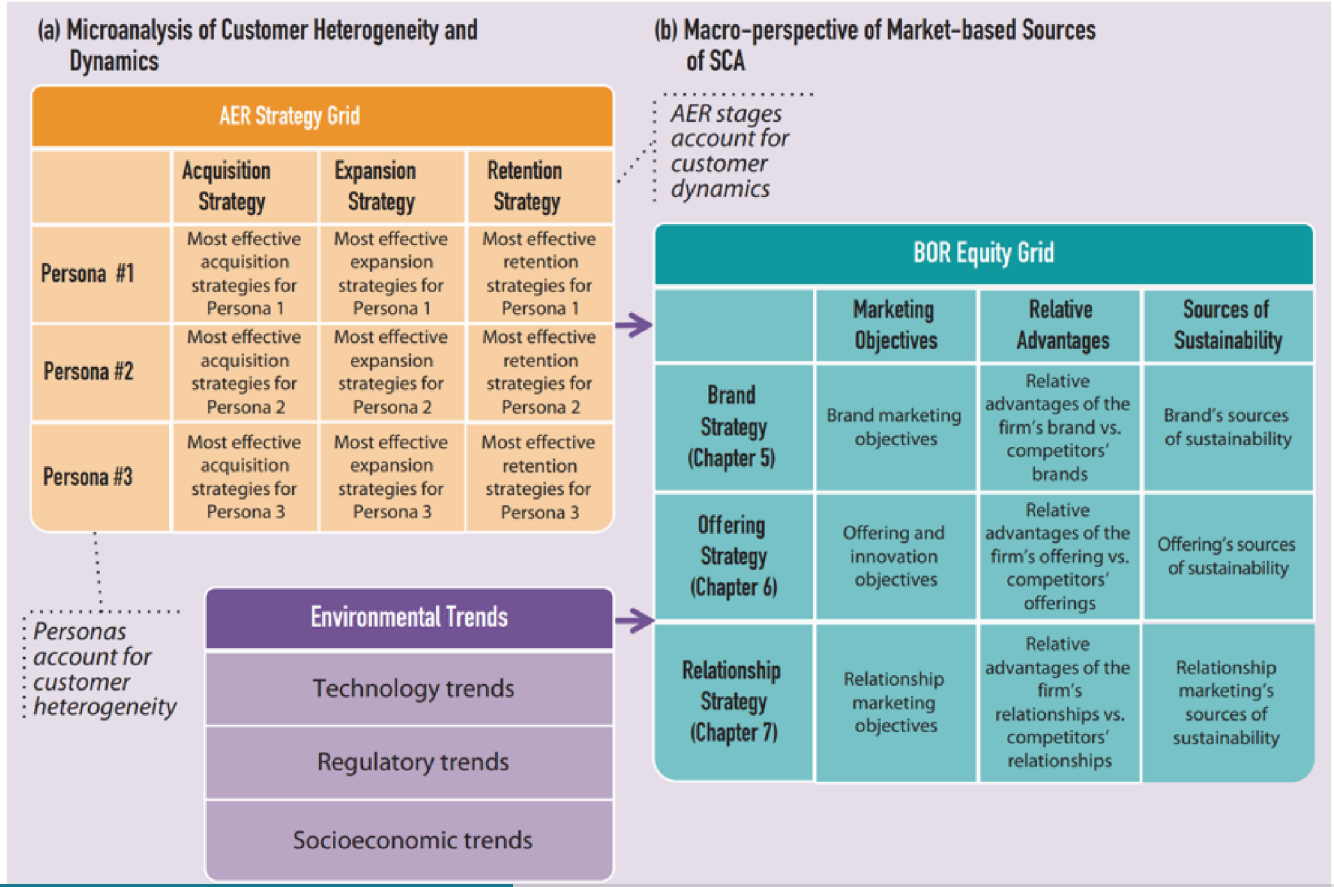
# Process for Managing Sustainable Competitive Advantage

- **Step 1: AER Strategy Grid** – each box in this grid describes the most effective strategy for a unique persona at a single point in time
- **Step 2: Key Trends** – managers should account for key environmental trends (e.g., long-term technology or regulatory changes) so that investments in BOR strategies lead to SCA, now and later
- **Step 3: BOR Equity Grid** – completed to describe three key pieces of information for the BOR strategies
  - → Marketing objectives
  - → Relative advantages
  - → Sources of sustainability
  - ⇉
- 

# Natural Sequence for Using Brands, Offerings, and Relationships to Build an SCA

- A firm should make brand decisions, which are influenced by the firm's overall positioning objectives from MP#1 and MP#2 and largely determine how the firm will be positioned in the overall marketplace and in existing customers' minds
- The firm can then focus on its offering decisions; product and service innovation and R&D efforts need to support both brand strategies and the firms positioning objectives
- Relationship strategies normally are determined last, because they involved the delivery and experiential aspect because in this case, boundary spanners are critical to the customer experience and the firm's overall value proposition
-

# Connections Between AER Strategy and BOR Equity Grids



# AER Strategy Grid: Micro-Analysis of Customer Heterogeneity and Dynamics

| AER Strategy Grid | | | |
|---|---|---|---|
| | **Acquisition Strategy** | **Expansion Strategy** | **Retention Strategy** |
| **Persona #1** | Most effective acquisition strategies for Persona 1 | Most effective expansion strategies for Persona 1 | Most effective retention strategies for Persona 1 |
| **Persona #2** | Most effective acquisition strategies for Persona 2 | Most effective expansion strategies for Persona 2 | Most effective retention strategies for Persona 2 |
| **Persona #3** | Most effective acquisition strategies for Persona 3 | Most effective expansion strategies for Persona 3 | Most effective retention strategies for Persona 3 |

- Output of MP#2 is a microanalysis of customer heterogeneity and dynamics in the firm's customer portfolio
- The insights from MP#2 then can be inserted into the AER strategy grid to reveal high-impact strategies
- Each box in this grid describes the most effective strategy for a unique persona at a single point in time

# BOR Equity Grid: Macro-Perspective of Market-Based Sources of SCA

**BOR Equity Grid**

| | Marketing Objectives | Relative Advantages | Sources of Sustainability |
|---|---|---|---|
| **Brand Strategy (Chapter 5)** | Brand marketing objectives | Relative advantages of the firm's brand vs. competitors' brands | Brand's sources of sustainability |
| **Offering Strategy (Chapter 6)** | Offering and innovation objectives | Relative advantages of the firm's offering vs. competitors' offerings | Offering's sources of sustainability |
| **Relationship Strategy (Chapter 7)** | Relationship marketing objectives | Relative advantages of the firm's relationships vs. competitors' relationships | Relationship marketing's sources of sustainability |

- Brand, Offering, and Relationship (BOR) Equity Grid represents last step in the process
  - → BOR equity grid offers a high-level summary of what the firm is trying to achieve with each BOR strategy
- Using the inputs from the AER strategy grid and key environmental trends, the BOR equity grid can be completed to describe three key pieces of information for the BOR strategies:
  - → Marketing objectives
  - → Relative advantages over competition
  - → Maintaining the SCA over time

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# AI Frontier: Identifying Market Structure from Social Engagement

Traditional market maps begin with predefined categories. An embedding approach lets observed consumer engagement reveal which brands occupy nearby competitive positions—even across category boundaries.

- Which brands attract overlapping audiences?
- Which rivals or complements sit outside the manager's existing category definition?
- How does proximity change after a major strategic event?

**Category view**

Competitors are defined first

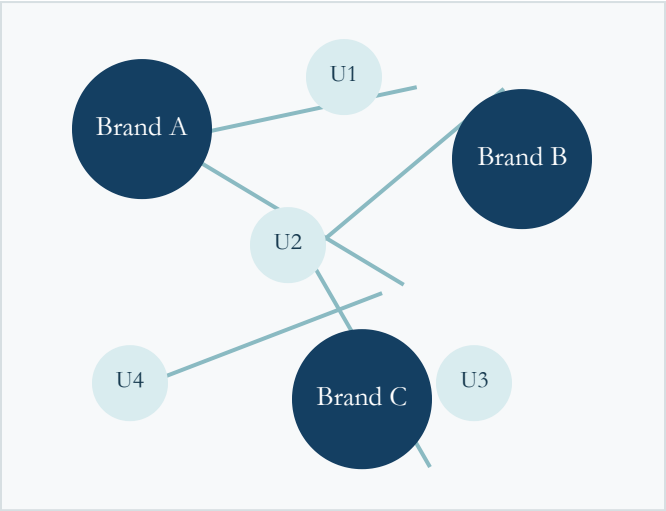
→

**Embedding view**

Competitors emerge from behavior

Yang, Zhang, and Kannan (2021), Journal of Marketing, 86(4), 37–56.

# Data: A Brand–User Engagement Network




**5,478**
brands

**25**
categories

**106.6M**
public engagement activities

**26.0M**
unique users

Facebook public fan-page engagement observed from January 2017 to January 2018; the study uses user identifiers and public engagement, not personal-profile content.

Source: Yang, Zhang, and Kannan (2021). Counts rounded from the published study.

# Method: From Engagement to a Market Map

01

**Build network**

Represent brands and users as nodes connected by public engagement.

→

02

**Learn embeddings**

A deep autoencoder compresses the sparse network into a 300-dimensional vector for each brand.

→

03

**Measure proximity**

Cosine similarity reveals brands with comparable engagement neighborhoods.

→

04

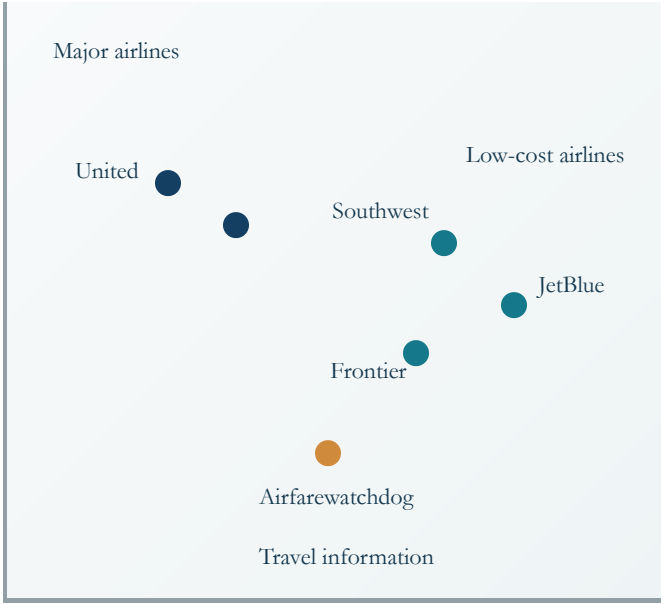
**Visualize structure**

t-SNE projects brand vectors into two dimensions for managerial interpretation.

| Market boundaries | Proximal brands | Substitutes and complements | Change over time |
|---|---|---|---|

Source: Yang, Zhang, and Kannan (2021). The 2D map is a visualization of the 300-dimensional representation.

# What the Embedding Reveals




## Within-category structure

Southwest sits near JetBlue, Frontier, and Allegiant; United aligns more closely with major domestic and international carriers.

## Cross-category adjacency

Airfarewatchdog appears near airlines even though a conventional industry taxonomy may place it elsewhere.

## Managerial interpretation

Proximity indicates shared engagement neighborhoods—not automatically direct substitution or causality.

Illustrative editable map based on examples reported by Yang, Zhang, and Kannan (2021); positions are schematic, not the paper's original coordinates.

# Dynamic Market Structure: Two Strategic Events

## Amazon acquires Whole Foods

| | |
|---|---|
| Amazon ↔ Whole Foods | +0.202 |
| Amazon ↔ Kroger | +0.165 |
| Amazon ↔ Lowe's | −0.184 |

The event draws Amazon closer to grocery brands and away from a home-improvement retailer in the learned representation.

## Tesla launches Model 3

| | |
|---|---|
| Tesla ↔ Maserati | −0.209 |

Premium niche → Broader auto set

Tesla moves closer to more mainstream brands such as Kia and Mazda after the mass-market launch.

Event-study changes are descriptive shifts in embedding similarity. They do not, by themselves, establish a causal effect.

Source: Yang, Zhang, and Kannan (2021), reported similarity changes around the two events.

# Managerial Value—and Important Limits

## Use the method to…

- scan for emerging competitors beyond predefined categories;
- track how launches, acquisitions, or campaigns shift brand neighborhoods;
- combine behavioral structure with customer surveys and search data.

## Interpret with care

- one platform may not represent the full market;
- platform demographics and algorithms shape observed engagement;
- similarity is descriptive and must be validated before strategic action;
- privacy and governance remain central.

What managerial decision would change if a newly proximal brand were treated as a competitor rather than an adjacent brand?

Source: Yang, Zhang, and Kannan (2021).

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# Takeaways

- All competitors react. Firms must address competitive attacks by building and maintaining sustainable competitive advantages (SCA).
- Customers generate sales and profits; firms must protect them from constant attacks by competitors. Although SCA are critical, firms must first establish their differentiated position with a targeted customer group before building SCA around this position.
- All SCA must meet three conditions: (1) customers must care about what it offers, (2) the firm must do it better than competitors, and (3) it must be hard to duplicate.
- In marketing domains, the primary sources of SCA are brands, offerings (innovative products or services), and relationships (BOR). The strongest SCA use all three strategies in combination to reinforce the differentiated and targeted appeal of a firm to customers.

# Takeaways

- Experiments can reveal the causality of BOR investments, by randomly assigning customers to multiple groups with different BOR investment levels/designs, including both treatment and control groups, to minimize potential confounds.
- Competitors have many ways to undermine a firm's SCA, including technical innovations, exploiting customers' changed desires, finding better solutions to a problem, and introducing "me-too" offerings with greater efficiency.
- A customer equity perspective implies that customers should be considered assets, managed and tracked that way, to improve firm performance. Investments in brands, offerings, and relationships represent important sources of customer equity.
- There are three inputs to the SCA framework: the output of MP#1 about what customers want and how the firm should position itself, the output of MP#2 about the most effective AER strategies as customers change, and long-term environmental trends.

# Takeaways

- The two outputs of the SCA framework are: a description of a firm's SCA now and in the future and a description of the BOR strategies needed to achieve it.
- The three-step process for managing SCA includes an AER Strategy Grid, an analysis of key environmental trends, and a BOR Equity Grid.
- Technology, regulatory, and socioeconomic trends constantly change; a firm's competitors constantly try to find new ways to satisfy customers' needs and desires. These changes all have the potential to disrupt any firm's market position.

# Readings

- ***Step-by-Step Guide to Smart Business Experiments*** (short non-technical review of using experiments in business with some good examples)
- ***Marketing Strategy Book***: Chapter 4